

Turkish Medicine API - Complete Documentation

Version: 2.0.0

Base URL: `http://localhost:3000`

Table of Contents

- [Overview](#)
- [Authentication](#)
- [Response Format](#)
- [Error Handling](#)
- [Endpoints](#)
 - [General Information](#)
 - [Medicine Data](#)
 - [Sheet-Specific Data](#)
 - [Search & Filter](#)
 - [Metadata & Statistics](#)
 - [Management](#)
- [Data Schema](#)
- [Examples](#)

Overview

The Turkish Medicine API provides programmatic access to the TITCK (Türkiye İlaç ve Tıbbi Cihaz Kurumu) medicine database. The API serves data from 5 different Excel sheets containing information about active, passive, and modified pharmaceutical products in Turkey.

Total Records: 17,713 medicine records

Data Sheets:

- Active Products: 7,887 records
- Passive Products: 9,759 records
- Products to be Deactivated: 61 records
- Newly Added Products: 3 records
- Modified Products: 3 records

Update Frequency: Weekly (every Monday at 9:00 AM by default)

Authentication

Currently, the API does not require authentication. All endpoints are publicly accessible.

Response Format

All responses are in JSON format with UTF-8 encoding.

Success Response

```
{
  "page": 1,
  "limit": 50,
  "total": 7887,
  "totalPages": 158,
  "data": [...]
}
```

Error Response

```
{
  "error": "Error message describing what went wrong"
}
```

HTTP Status Codes:

- 200 OK - Request successful
- 400 Bad Request - Invalid parameters
- 404 Not Found - Resource not found
- 500 Internal Server Error - Server error
- 503 Service Unavailable - Data not loaded yet

Error Handling

The API uses standard HTTP status codes to indicate success or failure.

Common Errors:

Status Code	Error Message	Description
400	Query parameter "q" is required	Search query missing
400	Parameters "field" and "value" are required	Filter parameters missing
404	Sheet not found	Invalid sheet name/slug
404	Medicine not found	Medicine ID doesn't exist
503	Data not loaded yet	Server starting or data loading failed

Endpoints

General Information

Get API Information

GET /

Returns API information, available endpoints, and sheet aliases.

Response:

```
{
  "name": "Turkish Medicine API",
  "version": "2.0.0",
  "description": "API for Turkish Medicine Database (TITCK) - All Sheets Supported",
  "endpoints": {
    "/": "API information",
    "/health": "Health check",
    ...
  },
  "sheets": [
    "AKTİF ÜRÜNLER LİSTESİ",
    "PASİF ÜRÜNLER LİSTESİ",
    ...
  ],
  "sheetAliases": {
    "active": "AKTİF ÜRÜNLER LİSTESİ",
    "passive": "PASİF ÜRÜNLER LİSTESİ",
    ...
  }
}
```

Health Check

```
GET /health
```

Returns server health status and data loading information.

Response:

```
{
  "status": "healthy",
  "dataLoaded": true,
  "recordCount": 17713,
  "lastLoadTime": "2025-10-18T06:30:30.569Z"
}
```

Medicine Data

Get All Medicines

```
GET /api/medicines
```

Returns all medicines from all sheets with pagination.

Query Parameters:

Parameter	Type	Default	Description
page	integer	1	Page number
limit	integer	50	Records per page

Example Request:

```
GET /api/medicines?page=2&limit=100
```

Response:

```
{
  "page": 2,
  "limit": 100,
  "total": 17713,
  "totalPages": 178,
  "data": [
    {
      "_sheet": "AKTİF ÜRÜNLER LİSTESİ",
      "İlaç Adı": "PAROL 500 MG TABLET",
      "la_ad": "PAROL 500 MG TABLET",
      "Barkod": 8699717690028,
      "ATC Kodu": "N02BE01",
      "atc_kodu": "N02BE01",
      "ATC Adı": "paracetamol",
      "atc_ad": "paracetamol",
      "Firma Adı": "ATABAY KİMYA SANAYİ VE TİC. A.Ş.",
      "firma_ad": "ATABAY KİMYA SANAYİ VE TİC. A.Ş.",
      "Reçete Türü": "Normal",
      "reete_tr": "Normal",
      "Durumu": "Aktif",
      "id": 101
    }
  ]
}
```

Get Medicine by ID

```
GET /api/medicines/:id
```

Returns a single medicine record by its ID.

Path Parameters:

Parameter	Type	Description
id	integer	Medicine ID

Example Request:

```
GET /api/medicines/100
```

Response:

```
{
  "_sheet": "AKTİF ÜRÜNLER LİSTESİ",
  "İlaç Adı": "CYLORIN 25 MG YUMUSAK KAPSUL (50 KAPSUL)",
  "İlaç Adı": "CYLORIN 25 MG YUMUSAK KAPSUL (50 KAPSUL)",
  "Barkod": 8680199196634,
  "ATC Kodu": "L04AD01",
  "atc_kodu": "L04AD01",
  "ATC Adı": "ciclosporin",
  "atc_ad": "ciclosporin",
  "id": 100
}
```

Error Response (404):

```
{
  "error": "Medicine not found"
}
```

Sheet-Specific Data

All sheet endpoints support pagination via `page` and `limit` query parameters.

List All Sheets

```
GET /api/sheets
```

Returns information about all available sheets.

Response:

```
{
  "totalSheets": 5,
  "sheets": {
    "AKTİF ÜRÜNLER LİSTESİ": {
      "slug": "active",
      "records": 7887,
      "endpoint": "/api/sheets/active"
    },
    "PASİF ÜRÜNLER LİSTESİ": {
      "slug": "passive",
      "records": 9759,
      "endpoint": "/api/sheets/passive"
    },
    ...
  }
}
```

Get Active Products

```
GET /api/sheets/active
```

Returns all active (currently marketed) pharmaceutical products.

Query Parameters: `page`, `limit`

Response: Same format as "Get All Medicines"

Get Passive Products

```
GET /api/sheets/passive
```

Returns all passive (discontinued) pharmaceutical products.

Query Parameters: `page`, `limit`

Get Products to be Deactivated

```
GET /api/sheets/to-be-deactivated
```

Returns products scheduled to be deactivated.

Query Parameters: `page`, `limit`

Get Newly Added Products

```
GET /api/sheets/newly-added
```

Returns recently added pharmaceutical products.

Query Parameters: `page`, `limit`

Get Modified Products

```
GET /api/sheets/modified
```

Returns products that have been recently modified.

Query Parameters: `page`, `limit`

Search & Filter

Search Medicines

```
GET /api/medicines/search
```

Search for medicines across all sheets or within a specific sheet.

Query Parameters:

Parameter	Type	Required	Description
q	string	Yes	Search query (case-insensitive)
sheet	string	No	Sheet name to search within
page	integer	No	Page number (default: 1)
limit	integer	No	Records per page (default: 50)

Example Requests:

```
# Search across all sheets
GET /api/medicines/search?q=parol

# Search within active products only
GET /api/medicines/search?q=parol&sheet=AKTİF%20ÜRÜNLER%20LİSTESİ

# Search with pagination
GET /api/medicines/search?q=aspirin&page=1&limit=10
```

Response:

```
{
  "query": "parol",
  "sheet": "all",
  "page": 1,
  "limit": 50,
  "total": 12,
  "totalPages": 1,
  "data": [...]
}
```

Filter by Field

```
GET /api/medicines/filter
```

Filter medicines by a specific field value.

Query Parameters:

Parameter	Type	Required	Description
field	string	Yes	Field name (Turkish or English)
value	string	Yes	Value to filter by (partial match)
sheet	string	No	Sheet name to filter within
page	integer	No	Page number (default: 1)
limit	integer	No	Records per page (default: 50)

Example Requests:

```
# Filter by barcode
GET /api/medicines/filter?field=Barkod&value=8699717

# Filter by company name
GET /api/medicines/filter?field=Firma%20Adı&value=BAYER

# Filter in specific sheet
GET /api/medicines/filter?field=Durumu&value=Aktif&sheet=AKTİF%20ÜRÜNLER%20LİSTESİ
```

Response:

```
{
  "field": "Barkod",
  "value": "8699717",
  "sheet": "all",
  "page": 1,
  "limit": 50,
  "total": 8,
  "totalPages": 1,
  "data": [...]
}
```

Metadata & Statistics

Get Available Columns

GET /api/columns

Returns all available column names.

Query Parameters:

Parameter	Type	Required	Description
sheet	string	No	Get columns for specific sheet

Example Requests:

```
# Get all columns
GET /api/columns

# Get columns for active products
GET /api/columns?sheet=AKTİF%20ÜRÜNLER%20LİSTESİ
```

Response:

```
{
  "sheet": "all",
  "columns": [
    "_sheet",
    "İlaç Adı",
    "la_ad",
    "Barkod",
    "ATC Kodu",
    "atc_kodu",
    "ATC Adı",
    "atc_ad",
    "Firma Adı",
    "firma_ad",
    "Reçete Türü",
    "reete_tr",
    "Durumu",
    "Açıklama",
    "aklama",
    "id"
  ]
}
```

Get Statistics

GET /api/stats

Returns statistics for all sheets.

Response:

```
{
  "totalRecords": 17713,
  "sheetCount": 5,
  "sheets": {
    "AKTİF ÜRÜNLER LİSTESİ": {
      "records": 7887,
      "columns": ["İlaç Adı", "Barkod", "ATC Kodu", ...]
    },
    "PASİF ÜRÜNLER LİSTESİ": {
      "records": 9759,
      "columns": ["İlaç Adı", "Barkod", "ATC Kodu", ...]
    },
    ...
  }
}
```

Get Download Metadata

```
GET /api/metadata
```

Returns information about the last data download.

Response:

```
{
  "downloadDate": "2025-10-18T06:24:41.856Z",
  "sourceDate": "E-Reçete İlaç Listesi",
  "sourceUrl": "https://titck.gov.tr/storage/Archive/2025/...",
  "filePath": "data/medicines.xlsx"
}
```

Management

Reload Data

```
POST /api/reload
```

Reloads data from the existing Excel file without downloading.

Response:

```
{
  "message": "Data reloaded successfully",
  "recordCount": 17713,
  "loadTime": "2025-10-18T06:30:30.569Z"
}
```

Download Latest Data

```
POST /api/download
```

Downloads the latest Excel file from TITCK and reloads data.

Response:


```
{
  "message": "File downloaded and data loaded successfully",
  "metadata": {
    "downloadDate": "2025-10-18T06:24:41.856Z",
    "sourceDate": "E-Reçete İlaç Listesi",
    "sourceUrl": "https://titck.gov.tr/storage/Archive/2025/...",
    "filePath": "data/medicines.xlsx"
  },
  "recordCount": 17713
}
```

Data Schema

Medicine Record Structure

Each medicine record contains both Turkish and English column names for flexibility.

Active Products Fields:

Turkish Column	English Column	Type	Description
_sheet	_sheet	string	Source sheet name
İlaç Adı	la_ad	string	Medicine name
Barkod	-	number	Barcode number
ATC Kodu	atc_kodu	string	ATC code
ATC Adı	atc_ad	string	ATC description
Firma Adı	firma_ad	string	Company name
Reçete Türü	reete_tr	string	Prescription type
Durumu	-	string	Status (Aktif/Pasif)
Açıklama	aklama	string	Notes/Description
Temel İlaç Listesi Durumu	temel_la_listesi_durumu	number	Essential medicines list status
Çocuk Temel İlaç Listesi Durumu	ocuk_temel_la_listesi_durumu	number	Children's essential medicines status
Yenidoğan Temel İlaç Listesi Durumu	yenidoan_temel_la_listesi_durumu	number	Newborn essential medicines status
Aktif Ürünler Listesine Alındığı Tarih	aktif_rnler_listesine_alnd_tarih	number	Date added to active list
id	id	number	Unique identifier

Passive Products Fields:

Similar to active products but with:

- Pasif Ürünler Listesine Alındığı Tarih (pasif_rnler_listesine_alnd_tarih) - Date added to passive list
- No essential medicines list fields

Examples

Example 1: Search for Pain Medications

```
curl "http://localhost:3000/api/medicines/search?q=parol&limit=5"
```

Response:

```
{
  "query": "parol",
  "sheet": "all",
  "page": 1,
  "limit": 5,
  "total": 12,
  "totalPages": 3,
  "data": [
    {
      "_sheet": "AKTİF ÜRÜNLER LİSTESİ",
      "İlaç Adı": "PAROL 500 MG TABLET",
      "Barkod": 8699717690028,
      "ATC Kodu": "N02BE01",
      "ATC Adı": "paracetamol",
      "Firma Adı": "ATABAY KİMYA SANAYİ VE TİC. A.Ş.",
      "id": 5500
    }
  ]
}
```

Example 2: Get Only Active Products from BAYER

```
curl "http://localhost:3000/api/medicines/filter?field=Firma%20Adı&value=BAYER&sheet=AKTİF%20ÜRÜNLER%20LİSTESİ&limit=3"
```

Example 3: Get Newly Added Medicines

```
curl "http://localhost:3000/api/sheets/newly-added"
```

Response:

```
{
  "sheet": "LİSTEYE YENİ EKLENEN ÜRÜNLER",
  "slug": "newly-added",
  "page": 1,
  "limit": 50,
  "total": 3,
  "totalPages": 1,
  "data": [
    {
      "_sheet": "LİSTEYE YENİ EKLENEN ÜRÜNLER",
      "İlaç Adı": "LYPTIMZIA 284MG/ 1,5 ML ENJEKSİYONLUK COZELTI",
      "Barkod": 8699074950520,
      "ATC Kodu": "C10AX16",
      "ATC Adı": "inclisiran",
      "Açıklama": "Ekleme yeni ürün/Aktif ürünler listesine eklendi.",
      "id": 1
    }
  ]
}
```

Example 4: Get Statistics

```
curl "http://localhost:3000/api/stats"
```

Example 5: Search Within Passive Products Only

```
curl "http://localhost:3000/api/medicines/search?q=aspirin&sheet=PASİF%20ÜRÜNLER%20LİSTESİ"
```

Example 6: Filter by Barcode Prefix

```
curl "http://localhost:3000/api/medicines/filter?field=Barkod&value=86997"
```

Rate Limiting

Currently, there is no rate limiting implemented. Consider implementing rate limiting in production environments.

CORS

CORS is enabled for all origins (*). Configure this appropriately for production use.

Support & Issues

For issues and feature requests, please refer to the project repository.

Data Source

All data is sourced from TITCK (Türkiye İlaç ve Tıbbi Cihaz Kurumu): <https://www.titck.gov.tr/dinamikmodul/43>

Data Update Schedule: Weekly, every Monday at 9:00 AM (configurable)

Version History

v2.0.0 (Current)

- Added multi-sheet support
- Clean English endpoint aliases
- Dual column names (Turkish + English)
- Enhanced search and filter with sheet selection
- Sheet-specific endpoints

v1.0.0

- Initial release
- Basic API functionality
- Single sheet support